Title : AI-Based Property Buy Worthiness Assessment and Property Value Forecasting Using Location, Market, Builder and Civic Data (Property Buy / Not Buy Recommender)

Aim : To develop a practical, privacy-preserving AI/ML decision-support system that evaluates whether a user-specified property in Jammu is worth buying and forecasts its potential future value. The system integrates public real-estate datasets, limited web-scraped civic and builder indicators, and a Retrieval-Augmented Generation (RAG) agent for transparent builder-credibility synthesis. It prioritises data-driven transparency, explicit bias mitigation, and client-side execution so that no user data leaves the device.

Introduction: Property purchase is one of the major financial decisions made by individuals and families. Unlike ordinary consumer purchases, evaluating a property requires simultaneous consideration of current price, historical price trends, location characteristics, infrastructure development, builder credibility, civic conditions and future growth potential. Existing real-estate platforms primarily facilitate property discovery by providing listings, prices, locations and basic characteristics. However, the information required to determine whether a specific property is actually worth buying is often distributed across different sources and difficult to evaluate collectively. Historical property prices may indicate market appreciation, infrastructure development may influence future demand, civic complaints can signal locality-level issues, and builder credibility can indicate project-related risk. The proposed project develops an AI/ML-based property decision-support system that evaluates a given property using multiple relevant indicators rather than simply recommending properties based on user preferences. The system focuses on property assessment rather than property recommendation: it evaluates a property provided by the user. All inference is performed client-side, ensuring that no user data leaves the device.

Abstract Property purchase is a high-stakes financial decision requiring simultaneous evaluation of price, market history, locality conditions, legal status, civic risk and builder credibility, yet in Jammu this information is scattered across unconnected sources. This project proposes an AI-Based Property Buy-Worthiness Assessment and Property Value Forecasting System that evaluates a user-specified property rather than recommending listings. The system uses only public data: (i) publicly displayed Jammu listings on 99acres (1,200 records, 36 localities) for prices and property attributes; (ii) Jammu Development Authority (JDA) layout approvals and master-plan records for approval status and land use; (iii) J&K RERA registers for project registration, builder history and complaints; (iv) Jammu Municipal Corporation and Smart City Jammu records for civic complaints, water, sewage and road data; (v) J&K Revenue and Registration Department records and circle rates for ownership and registry type; (vi) J&K SDMA flood-hazard maps; and (vii) OpenStreetMap for distances, schools, hospitals and transport. A local Retrieval-Augmented Generation (RAG) component synthesises builder credibility from public RERA and news documents. These sources yield 35 variables (28 base, 7 engineered). An ensemble of Random Forest, XGBoost and LightGBM forecasts 1-, 3- and 5-year values. Buy-worthiness is a transparent weighted composite: Score = $100 \times \Sigma\, w \cdot I$, where each $I \in [0,1]$ is a min–max-normalised index built from related variables: valuation fairness (25%; price-versus-locality and price-versus-type ratios, locality price rank), growth outlook (25%; past growth, forecast growth, growth momentum), legal clarity (20%), builder credibility (10%), location and amenity access (10%), and inverted civic risk (10%). Scores ≥70 indicate Buy, 40–69 Maybe and <40 No-Buy; missing JDA approval or an unclear registry caps the verdict at Maybe. Initial weights are expert-defined, then validated against SHAP

importances and stratified cross-validated performance, while a classifier supplies the final label with confidence. Explainable AI shows each factor's contribution to every verdict, so buyers can see why a property is rated as it is. Evaluation uses R^2, MAE, RMSE, MAPE, accuracy and F1, with bias audits across localities and price bands. All inference runs client-side, so no user data leaves the device, in line with the V.E.T.S. framework.